

Supporting Information for ”Poleward shift of moisture sources over Greenland driven by enhanced Arctic warming”

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Introduction

The Supporting Information contains text describing the VSD method for moisture tracking and how it compares to Lagrangian trajectory methods. It also contains additional figures (Figures S1 to S5) to show the spatial coverage of the Greenland quadrants, the average moisture source fraction for the Boreal Spring and Autumn seasons, how Greenland moisture sources are simulated using a free-running GISS ModelE2.1 simulation, how Northwest quadrant moisture source trends vary with season, and how Northwest quadrant moisture source fractions and surface evaporation change with time using ten-year averages as opposed to five-year averages, for both nudged and free-running GISS simulations.

Text S1. Water Vapor Source Distribution (VSD) tracers are a type of painted water tracer experiment long employed by GISS (e.g. Koster et al., 1986). The "painted water" method adds an additional set of tracers to the model which experience all of the same processes as regular model water, but which have no influence on the modeled climate itself. This method can track the sources of both precipitation and water vapor (Nusbaumer & Noone, 2018), although for this study we chose to focus solely on precipitation, given its impact on Greenland SMB and ice sheet isotope ratios. Here the painted regions are generalized using global basis functions, which avoids defining regions a priori (Kelley, 2003). We have previously deployed this method for paleoclimate simulations (Lewis et al., 2010, 2013, 2014), but here we use it in a modern-climate context. Unlike Lagrangian techniques, this method can only be applied to models that already have numerical water tracers, and thus tend to have a lower spatial and temporal resolution. However, the advantage is that the tracers include all atmospheric processes, such as rain re-evaporation and mixing, and have no residence time requirement, allowing for them to capture moisture sources that would otherwise be missed using trajectory methods (Sodemann et al., 2008a; Nusbaumer & Noone, 2018).

We apply 144 VSD tracers distributed using spherical harmonics as basis functions, resolving wave number 11 and providing a functional vapor source distribution of 8° latitude by 10° longitude. These basis functions are orthogonal to one another, allowing the VSD to be determined by the weighted sum of these 144 tracers. They enable a computational cost savings of approximately 6x given the 828 tracers required to explicitly

paint each $8^{\circ} \times 10^{\circ}$ grid box over the earth's surface. Here we set the tracers to initialize at the point of surface evaporation, allowing for evaporation over all surface types.

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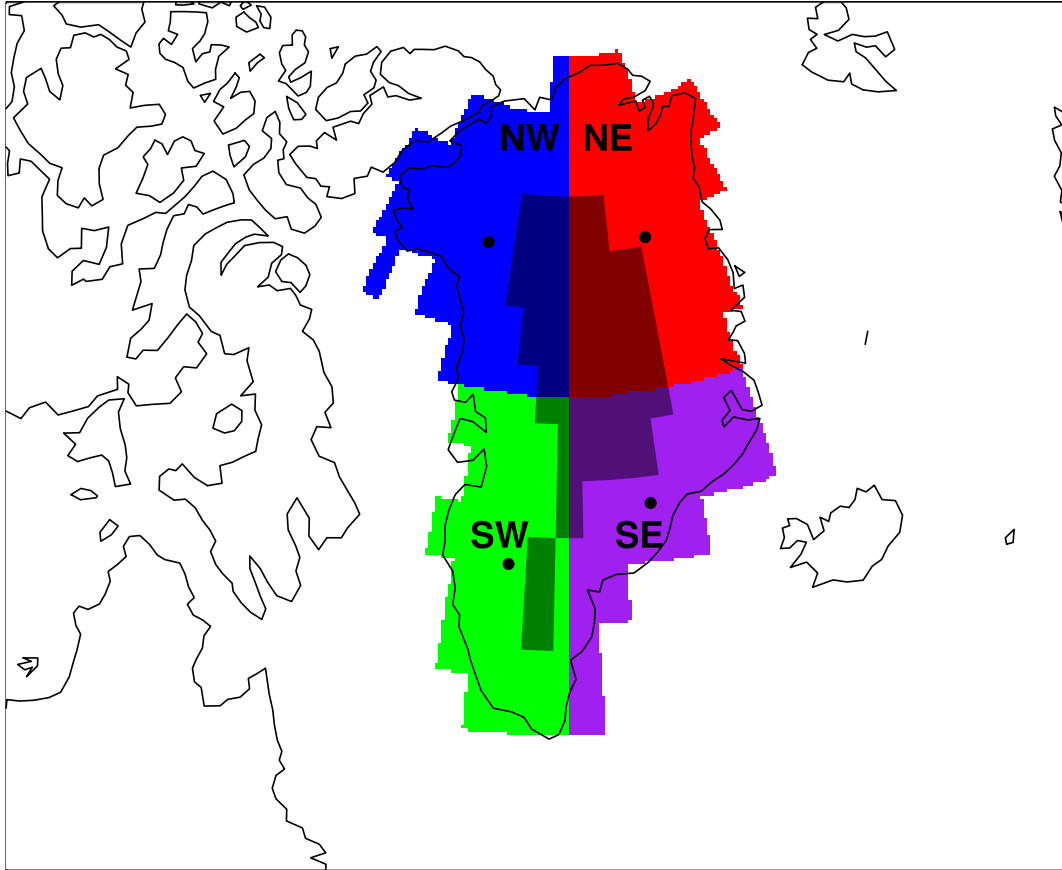


Figure S1. Map of the four different Greenland quadrants: Northwest (NW-blue), Northeast (NE-red), Southwest (SW-green), and Southeast (SE-purple). The grey-shading represents grid-cells with a surface altitude above 2000 meters. The black dots in each quadrant represent the area-weighted latitude/longitude center of each respective quadrant.

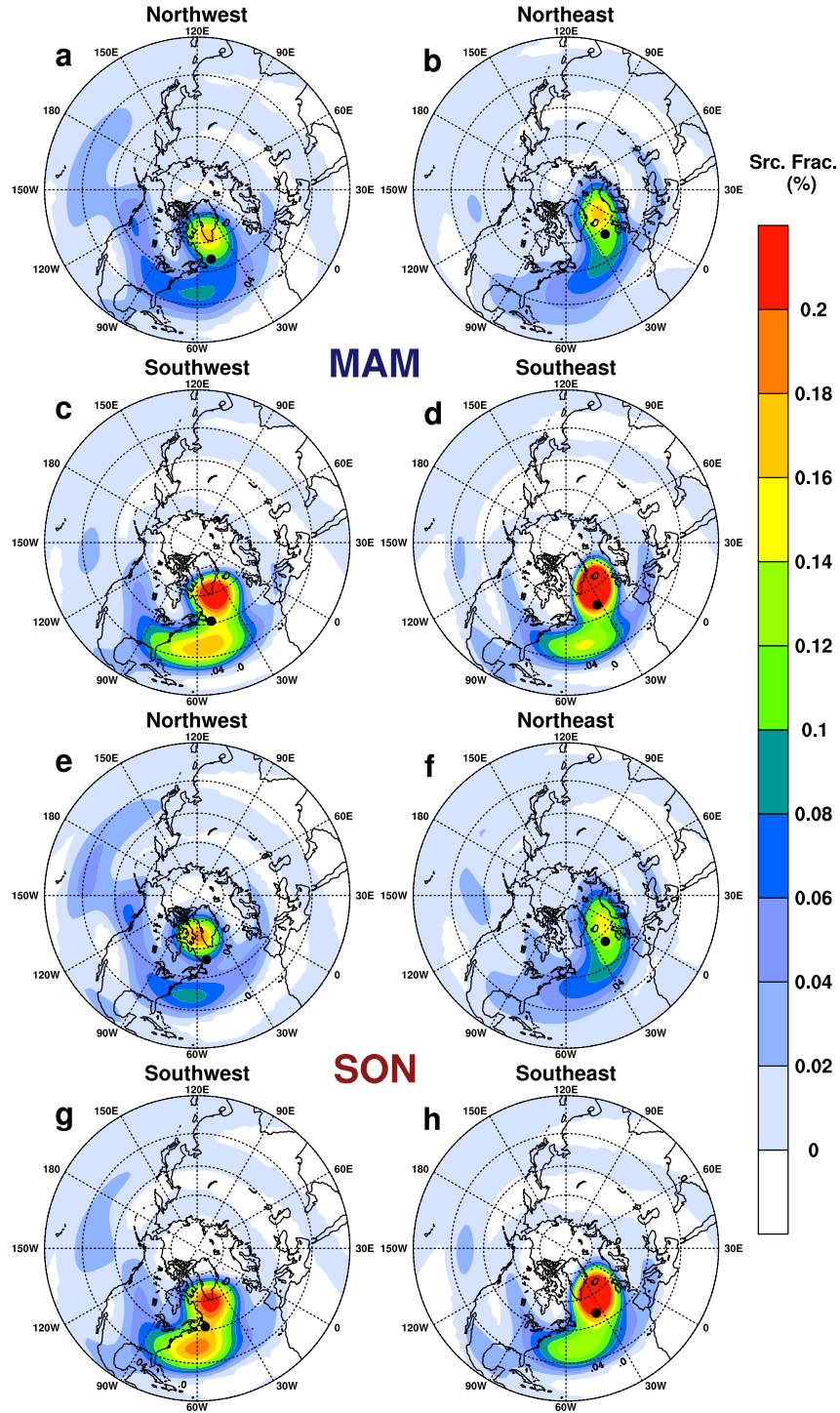


Figure S2. Climatological moisture source fraction in percent (%) averaged over March-April-May (MAM) for Northwest (a), Northeast (b), Southwest (c), and Southeast (d) Greenland, in the MERRA2-nudged simulation. The September-October-November (SON) average moisture source is also shown for Northwest (e), Northeast (f), Southwest (g), and Southeast (h) Greenland. The moisture fraction is defined here as the fraction of total precipitation for a given Greenland quadrant that is sourced from each grid box. The black dot in each plot represents the source-weighted latitude/longitude center of each respective source distribution.

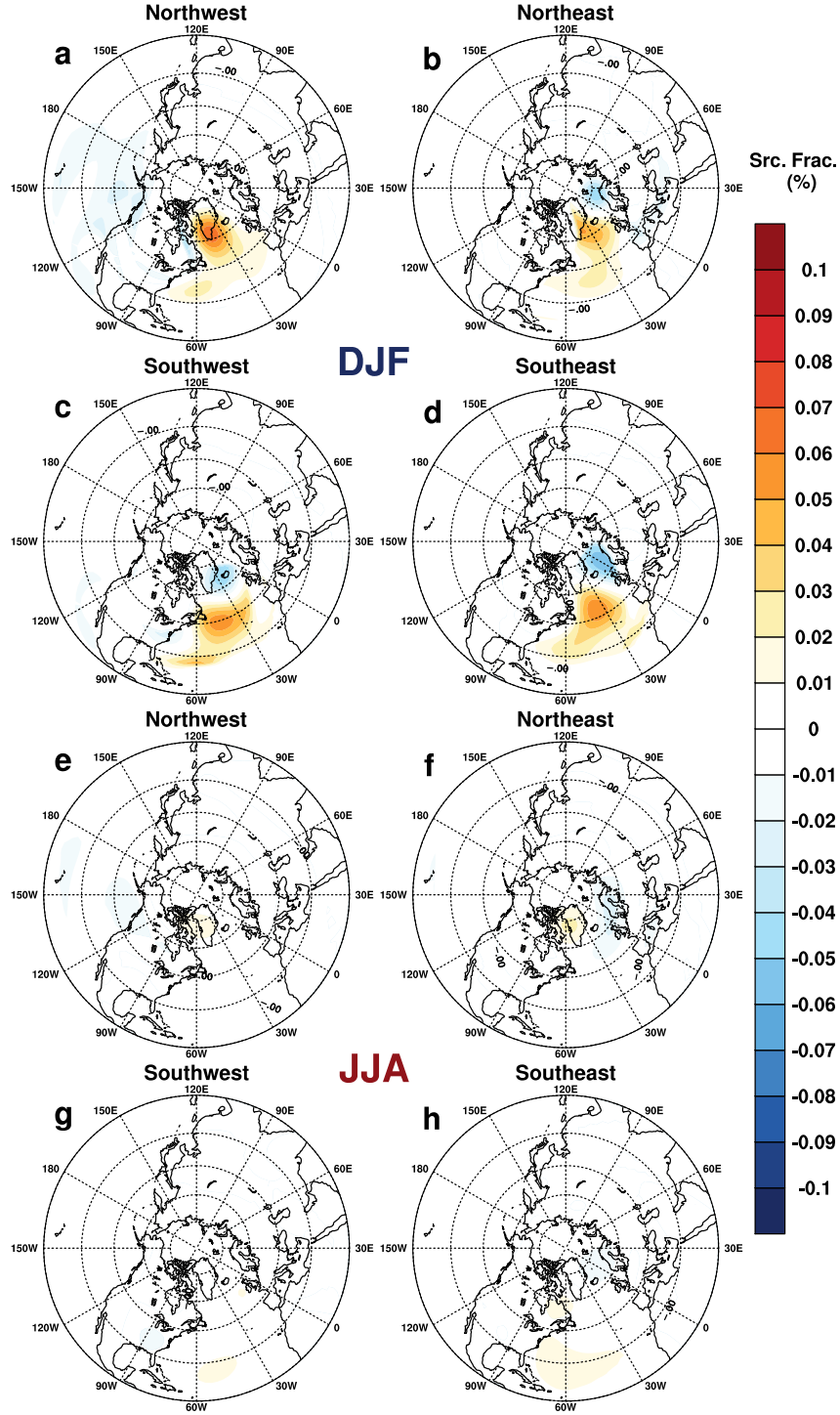


Figure S3. Difference in climatological moisture source fraction in percent (%) between the MERRA2-nudged simulation and free-running ensemble averaged over December-January-February (DJF) for Northwest (a), Northeast (b), Southwest (c), and Southeast (d) Greenland. The June-July-August (JJA) average moisture source difference is also shown for Northwest (e), Northeast (f), Southwest (g), and Southeast (h) Greenland. Orange colors indicate a larger moisture source fraction for the nudged simulation, while blue colors indicate a larger moisture source fraction for the free-running simulations.

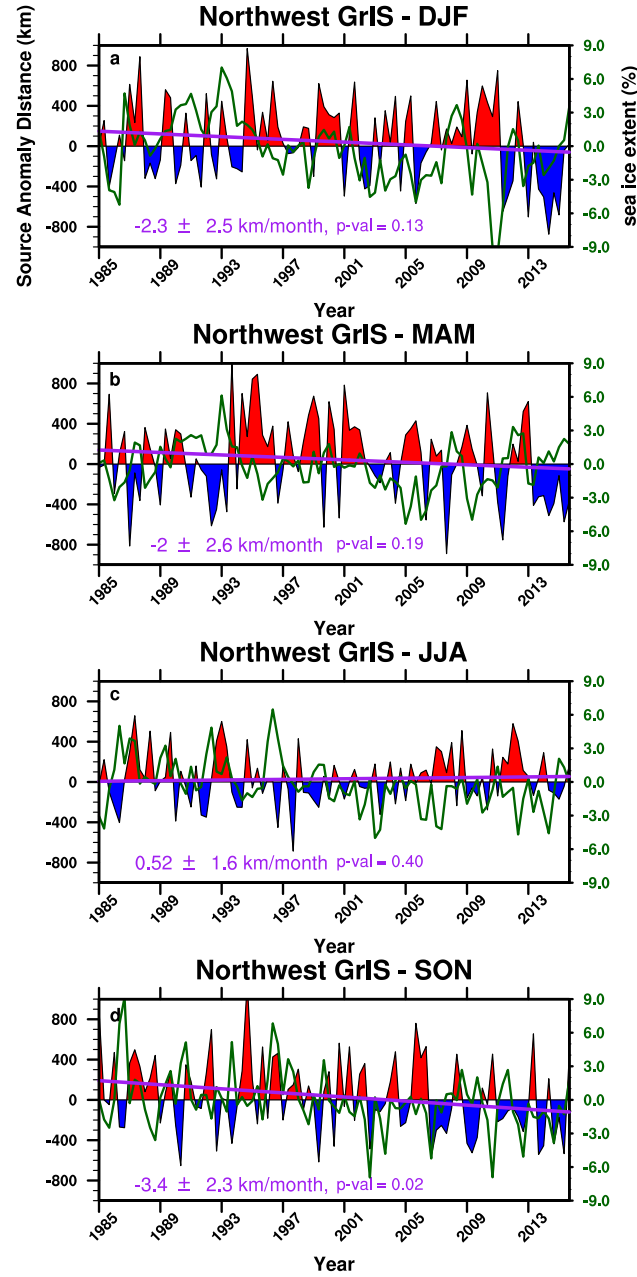


Figure S4. Time series of December-January-February (a), March-April-May (b), June-July-August (c), and September-October-November (d) monthly anomalies in distance between moisture source center and the Northwest Greenland quadrant center for the years 1985 to 2015 in the MERRA2-nudged simulation. Red colors indicate distances that are longer than the monthly average, while blue colors indicate distances that are shorter. Also plotted is the long-term trend for each season as calculated using a simple linear regression between time and the source distance, with the slope shown in km per month (purple). The green line shows the anomalous sea ice extent in percent for the Baffin Bay region (60° N - 80° N, 45° W - 75° W).

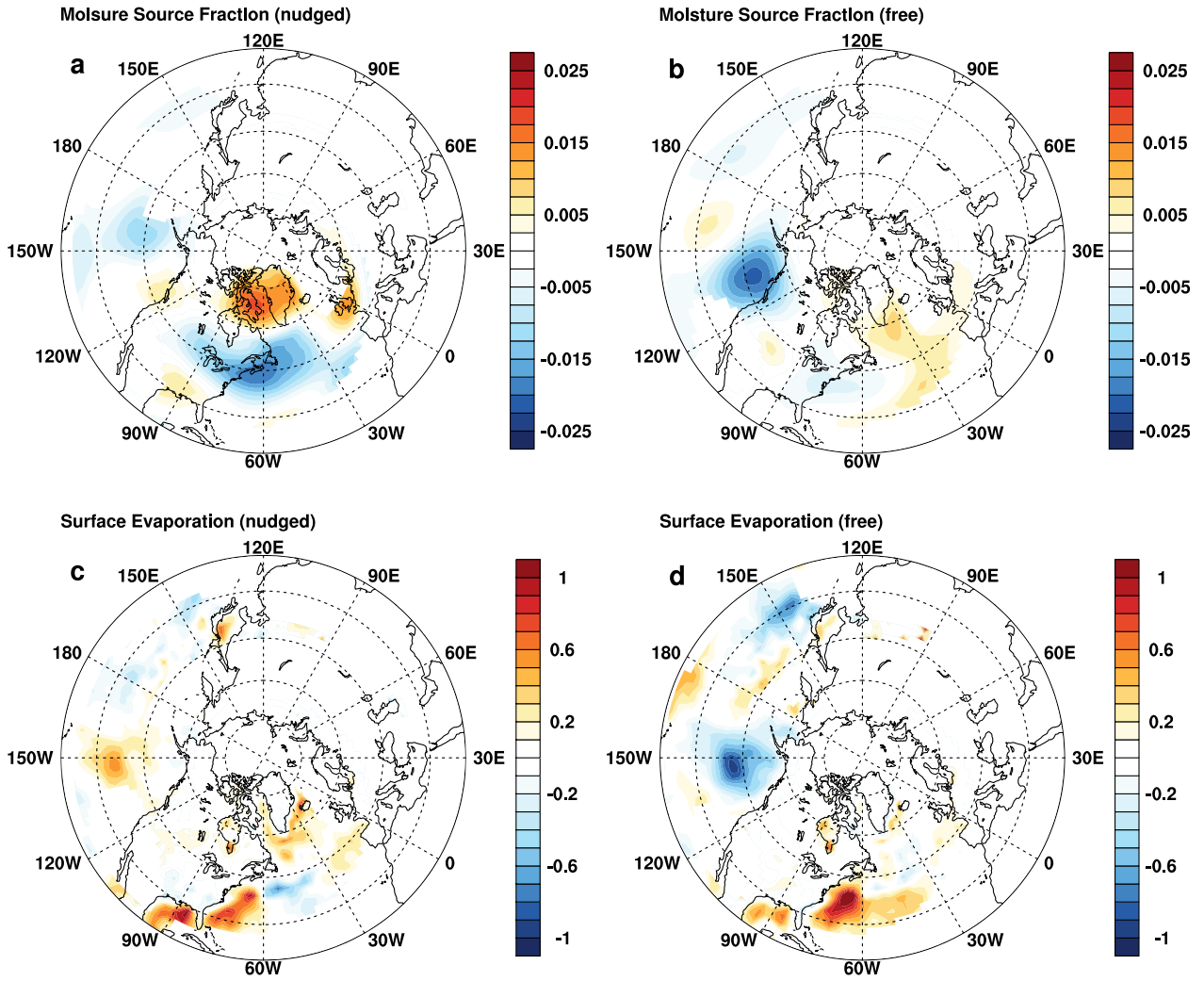


Figure S5. Difference in moisture source fraction between the first ten September-October-November seasons (1985-1995) and the last ten September-October-November seasons (2005-2015) of the MERRA2-nudged simulation (a), and the ensemble average of the free-running simulations (b), for the Northwest Greenland quadrant. Also shown are the differences in surface evaporation in mm/day over that same time period for the nudged (c) and free-running (d) simulations. Orange colors indicate a larger moisture source fraction or evaporation flux over time, while blue colors indicate a smaller fraction or flux. Difference values are also masked to only show changes in regions with an average moisture source fraction of 0.01 % or higher.